

Attention is All You Need

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, 30. [github link](#)

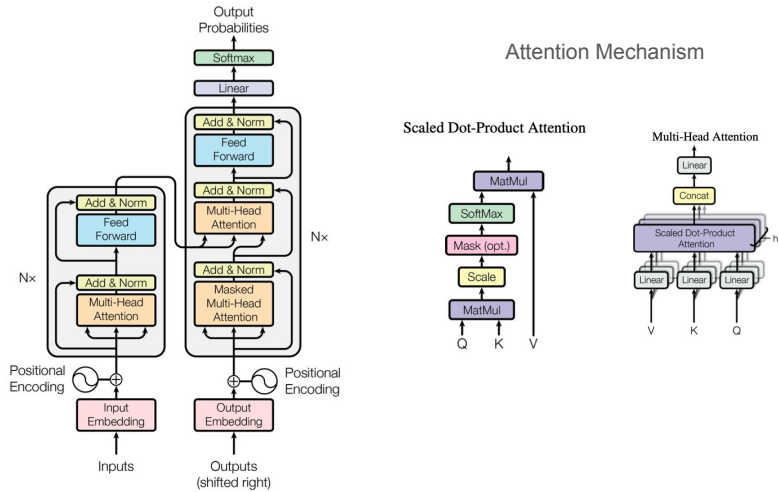
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Introduction

Problem: RNNs process data sequentially, creating a computational bottleneck that prevents parallelization.

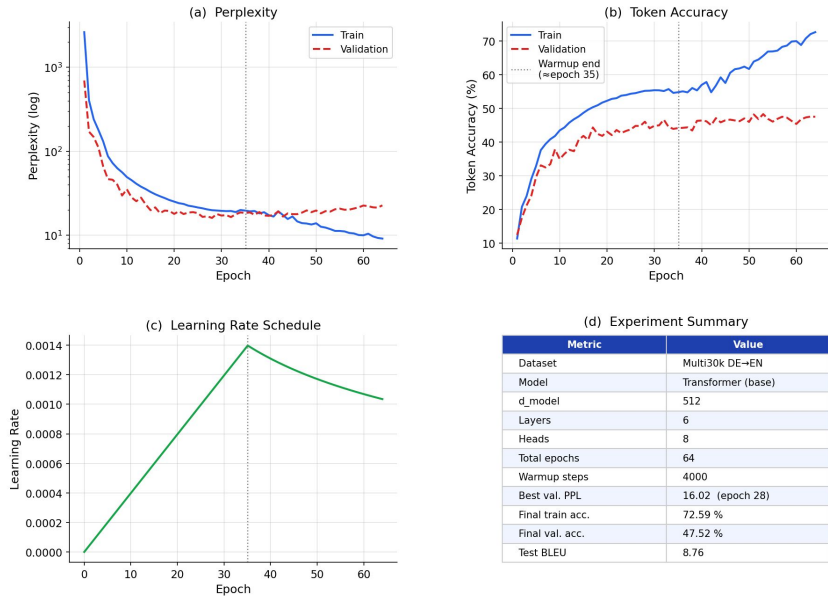
Solution: The Transformer replaces RNNs with **Self-Attention**, analyzing the entire sequence simultaneously to unlock massive parallel compute.

Model Architecture



Results

Transformer DE→EN Translation — Training Report



Challenges

Outdated dependencies, torchtext API removed, No MPS GPU support

Dataset - Multi30k
Hardware - M4 Pro (MPS) (Apple Silicon)
Optimizer, Regularization, Residual Dropout, Label Smoothing
Task: German → English neural machine translation on the Multi30k dataset

Training Setup